ABDOMINAL PAIN

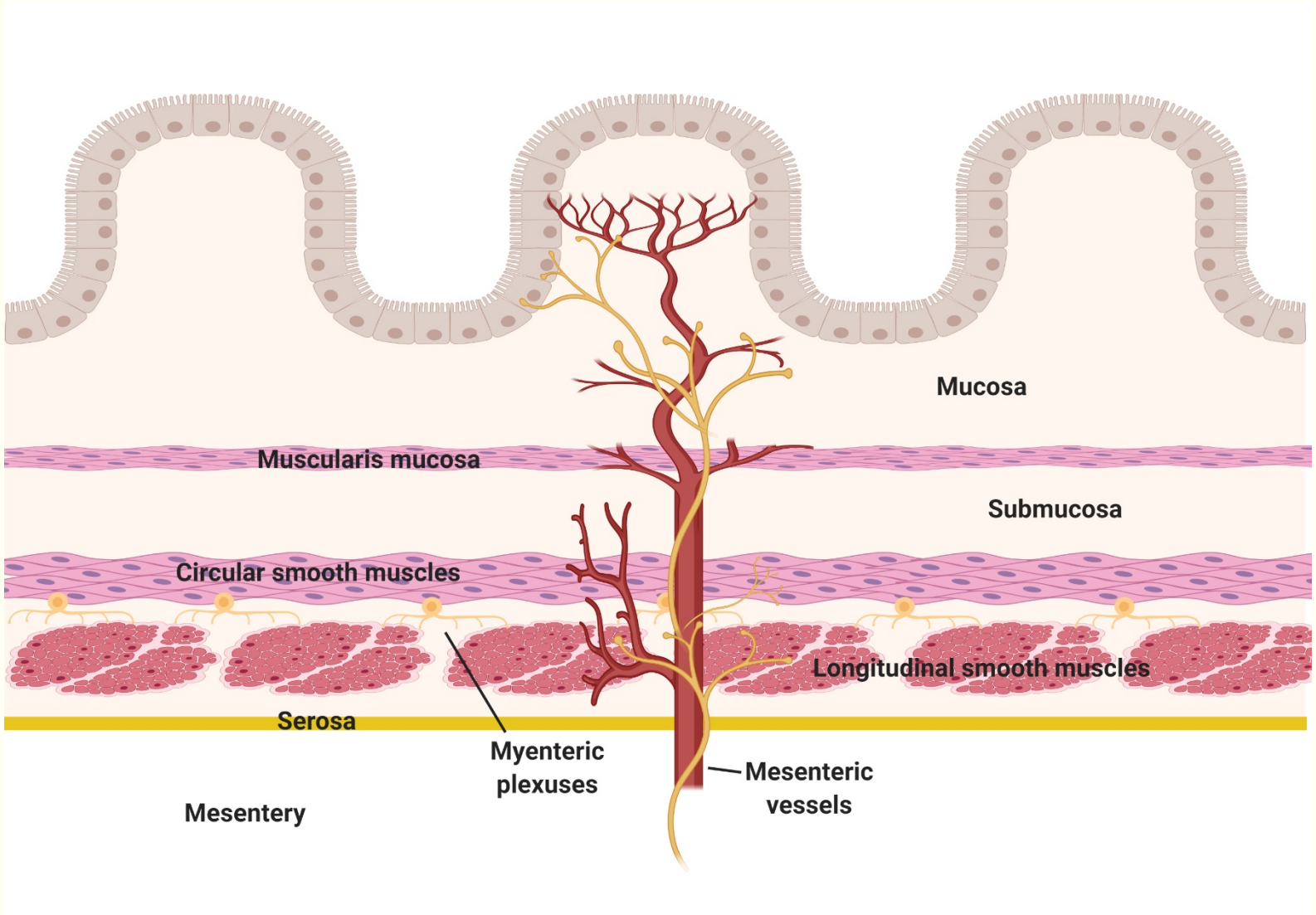
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Abdominal pain

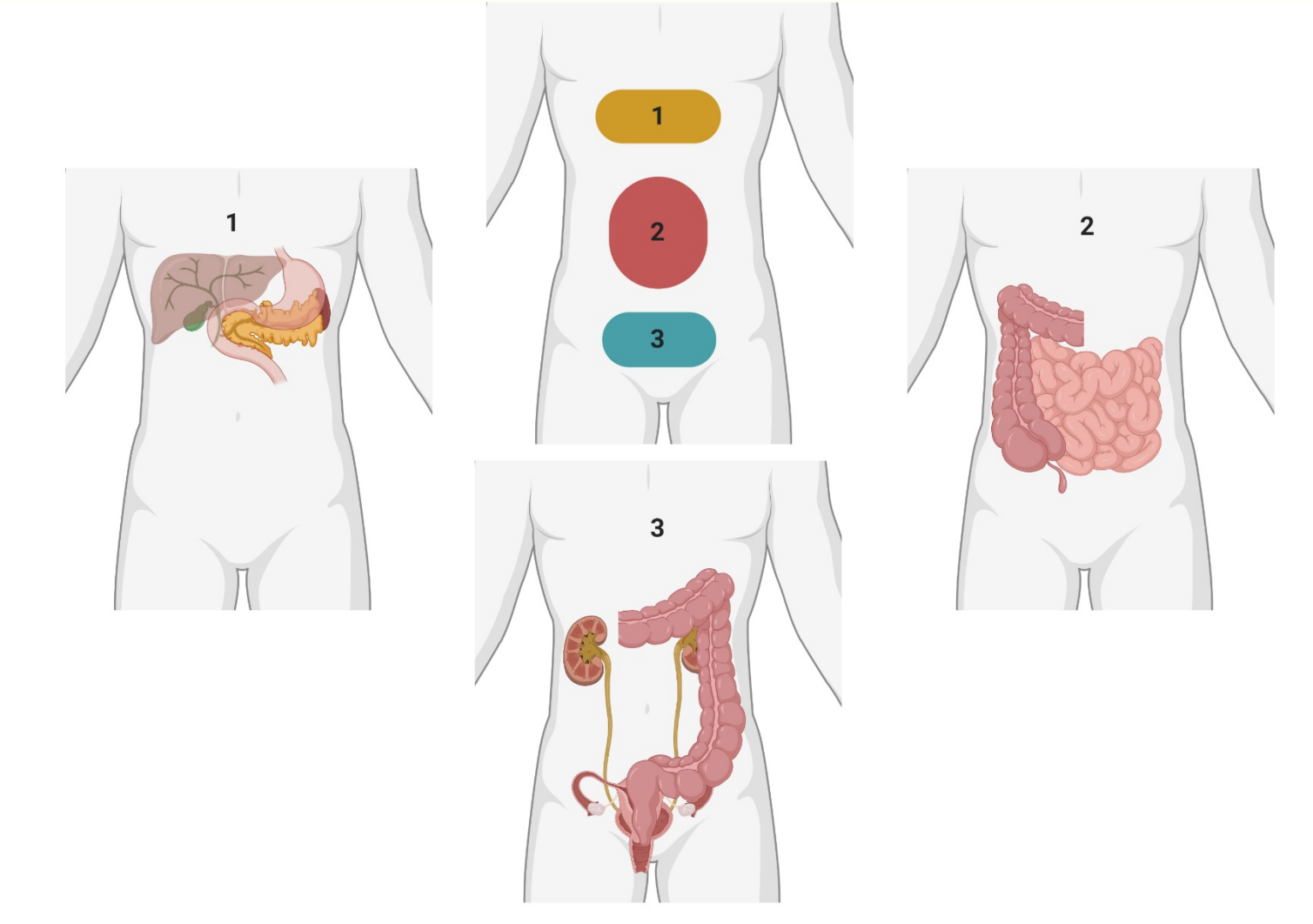
- **Visceral pain**: vague in onset and localization, dull-aching
- **Somatic-parietal pain**: more intense, sharp, and localized
- **Referred pain**: perceived at a point distant from the inciting pathology
- Pain from the abdominal viscera is primarily the result of stretch and distention, transduced by mechanoreceptor and inflammatory mediator
- Cutting and burns will not be perceived as noxious

Sensory innovation of GI tract

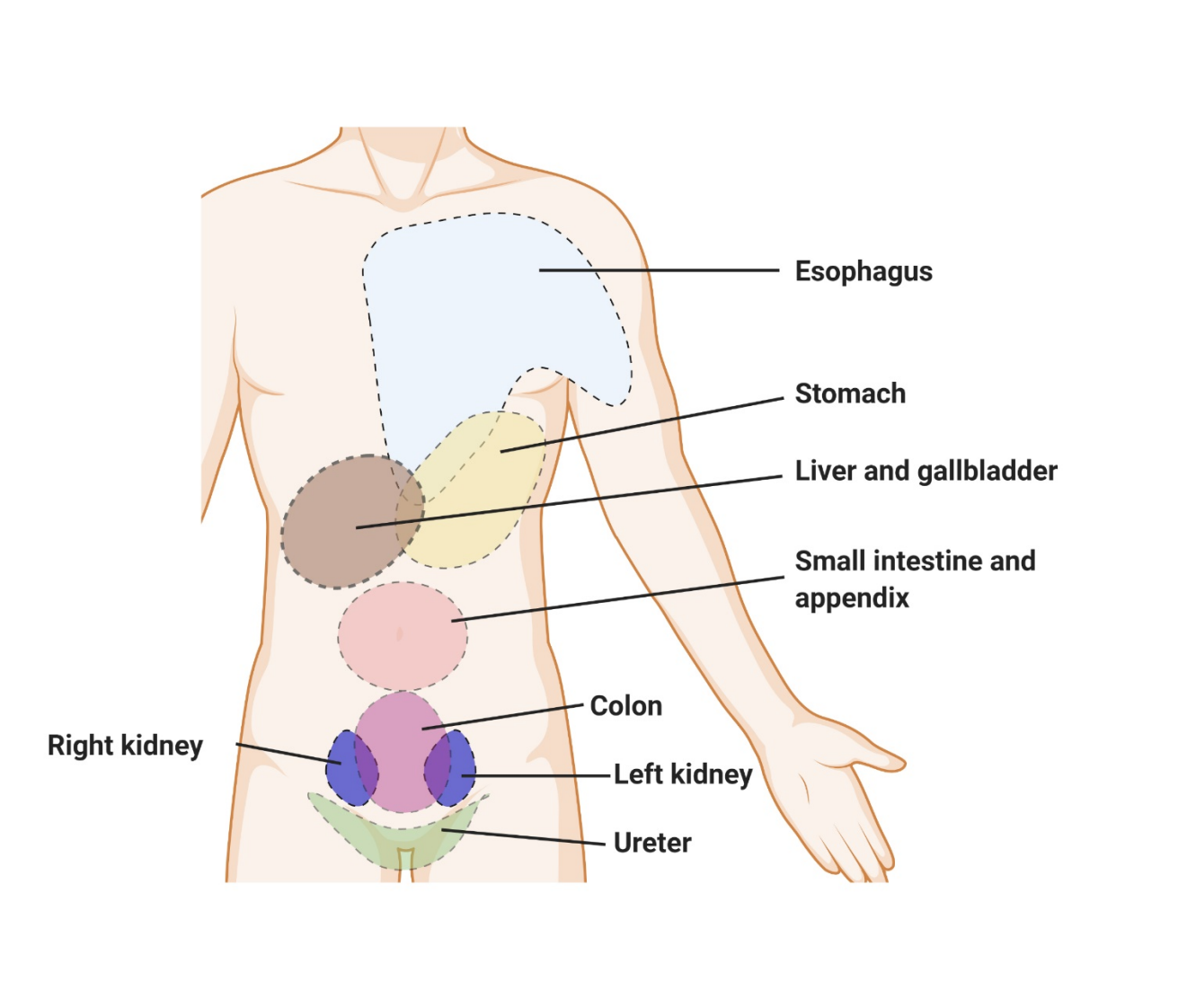


- Mesentery
- Peri-vascular
- Serosal
- Intra-muscular
- Intra-ganglionic
- Mucosa

Visceral pain



Surface area of visceral pain from different organs

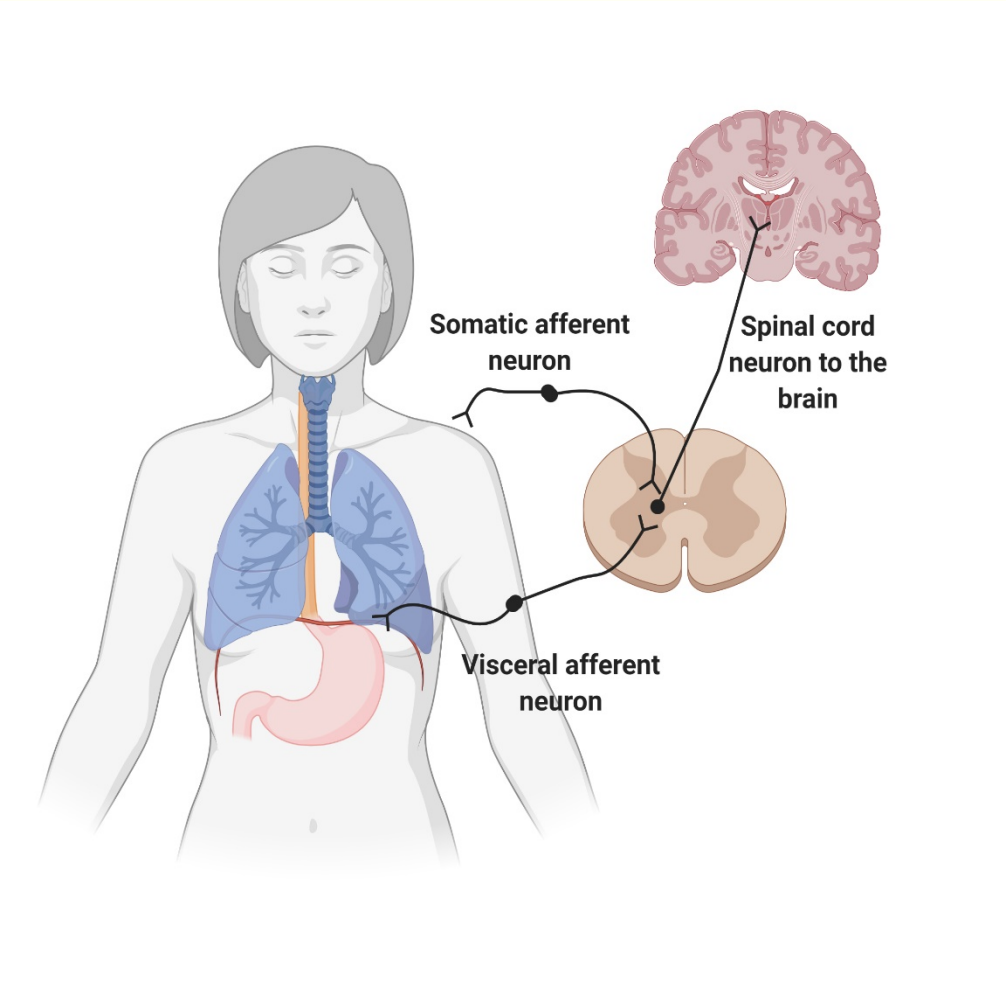


Somatic-parietal pain

- Mediated by A-delta fibers
- Conveys through somatic spinal nerves
- Reach the spinal cord in the peripheral nerves that correspond to the cutaneous dermatomes of the skin from T6 –L1
- Arising from noxious stimulation of the parietal peritoneum
- Sharp, sudden, well-localized pain
- Usually aggravated by movement or vibration

Referred pain

- Pain is felt in areas remote from the diseased organ
- Visceral afferent neurons and somatic afferent neurons from a different anatomic region converge on second-order neurons in the spinal cord at the same spinal segment.



Mechanisms of abdominal pain

- **Inflammation of the Parietal Peritoneum**: irritation from gastric acid (GU perforation), bacterial contamination (gut perforation, PID), pancreatic juice (pancreatitis), blood (hemoperitoneum)
- Accentuated by palpation or by movement → rebound tenderness
- Patients usually lie quietly in bed to avoid movement
- Tonic reflex spasm of the abdominal musculature, localized to the involved body segment → guarding

Mechanisms of abdominal pain

- **Obstruction of hollow viscera**:
- Intermittent or colicky abdominal pain that is not well localized
- The location depends on the organ
- Sudden distention of the biliary tree produces a steady rather than colicky type of pain (biliary colic is misleading)

Mechanisms of abdominal pain

- **Vascular Disturbances:**
- SMA thrombosis or occlusion: pain out of proportion to physical findings
- Rupture of AAA: can be sudden and catastrophic

Mechanisms of abdominal pain

- **Abdominal wall pain:**
- Usually constant and aching
- Worsen by movement, pressure, prolonged standing
- Such as muscle pain, abdominal wall hematoma

Mechanisms of abdominal pain

- **Metabolic causes:** DKA, porphyria, hereditary angioedema
- **Neurogenic causes:** herpes zoster, spinal nerve root compression
- **Functional causes**: IBS, functional dyspepsia

Common causes of acute abdominal pain

Cause	Onset	Location	Characteristics	Intensity
Appendicitis	Gradual	Periumbilical then RLQ	Aching	++
Cholecystitis	Acute	RUQ	Constricting	++
Pancreatitis	Acute	Epigastric/back	Boring	++ to +++
Diverticulitis	Gradual	LLQ	Aching	++ to +++
Perforated PU	Sudden	Epigastric then diffuse	Burning	+++
Small bowel obstruction	Gradual	Periumbilical/ diffuse	Cramping	++
Mesenteric ischemia	Sudden	Periumbilical/ diffuse	Agonizing	+++
Ruptured AAA	Sudden	Diffuse, back	Tearing	+++
Ruptured ectopic pregnancy	Sudden	Either LQ	Localizing	++

THANK YOU